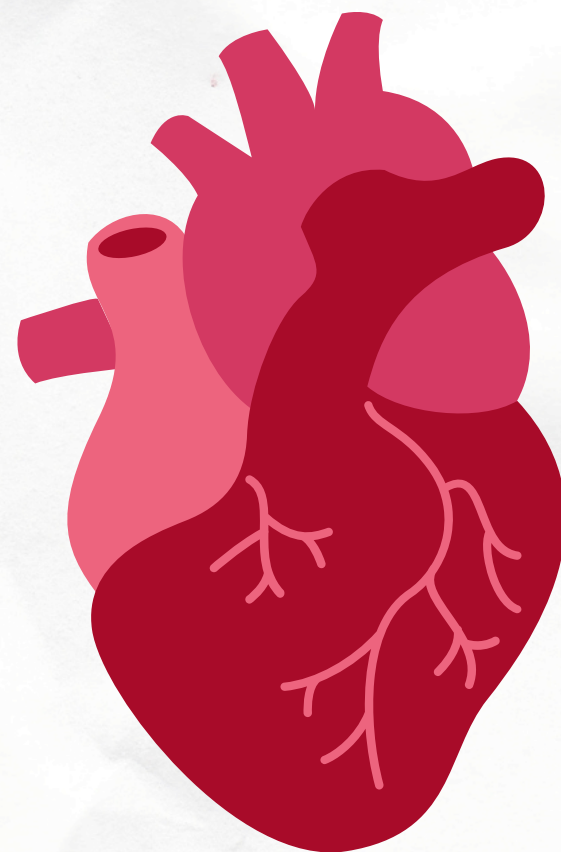
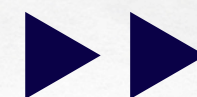




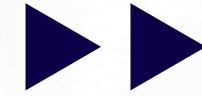
HEART DISEASE DIAGNOSTIC ANALYSIS



Ramkuvar Prajapati



➤ INTRODUCTION



- 
- A light blue curved arrow icon pointing downwards and to the right, located to the left of the list.
- Heart disease remains one of the leading causes of death globally, influenced by various factors including age, lifestyle, and genetics.
 - This project aims to analyze heart disease data to identify patterns and key risk factors associated with the condition.
 - Key variables explored include age, cholesterol levels, chest pain type, blood pressure, and heart rate.
 - The analysis is conducted using Python for data exploration and visual representation.
 - Insights from this project can aid healthcare professionals in identifying high-risk groups and understanding critical health metrics for better decision-making.





DETAILS OF DATA



- age: Age of the individual in years.
- sex: Gender of the individual (0 = Female, 1 = Male).
- cp (chest pain type): Type of chest pain experienced (0 = Typical angina, 1 = Atypical angina, 2 = Non-anginal pain, 3 = Asymptomatic).
- trestbps (resting blood pressure): Resting blood pressure in mm Hg on admission.
- chol (cholesterol): Serum cholesterol level in mg/dL.
- fbs (fasting blood sugar): Fasting blood sugar > 120 mg/dL (1 = True, 0 = False).
- restecg (resting electrocardiographic results): ECG results (0 = Normal, 1 = ST-T wave abnormality, 2 = Left ventricular hypertrophy).
- thalach (maximum heart rate achieved): Maximum heart rate achieved during exercise.
- exang (exercise-induced angina): Exercise-induced angina (1 = Yes, 0 = No).
- oldpeak: ST depression induced by exercise relative to rest.
- slope: Slope of the peak exercise ST segment (0 = Upsloping, 1 = Flat, 2 = Downsloping).
- ca (number of major vessels): Number of major vessels (0-3) colored by fluoroscopy.
- thal: Thalassemia (0 = Null, 1 = Fixed defect, 2 = Normal, 3 = Reversible defect).
- target: Presence of heart disease (1 = Disease, 0 = No disease).



```
df.head()
df.info()
df.describe()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1025 non-null   int64
1   sex         1025 non-null   int64
2   cp          1025 non-null   int64
3   trestbps    1025 non-null   int64
4   chol        1025 non-null   int64
5   fbs         1025 non-null   int64
6   restecg     1025 non-null   int64
7   thalach     1025 non-null   int64
8   exang       1025 non-null   int64
9   oldpeak     1025 non-null   float64
10  slope       1025 non-null   int64
11  ca          1025 non-null   int64
12  thal        1025 non-null   int64
13  target      1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```




DETAILS OF DATA



- **age:** Ranges from 29 to 77 years, with an average age of 54.
- **trestbps:** Resting blood pressure varies between 94 and 200 mm Hg, averaging 131 mm Hg.
- **chol:** Cholesterol levels span 126 to 564 mg/dL, with a mean of 246 mg/dL.
- **thalach:** Maximum heart rate achieved ranges from 71 to 202 bpm, averaging 149 bpm.
- **oldpeak:** ST depression ranges from 0.0 to 6.2, with an average of 1.07.
- **ca:** Major vessels colored by fluoroscopy range from 0 to 4, most commonly 0 or 1.

These statistics provide a snapshot of the dataset's patient demographics and health indicators.

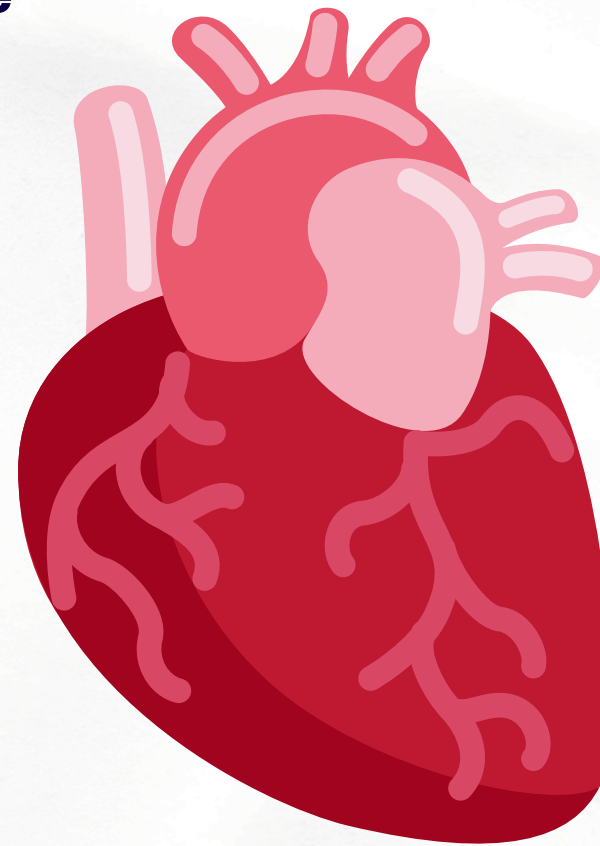
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	54.434146	0.695610	0.942439	131.611707	246.000000	0.149268	0.529756	149.114146	0.336585	1.071512	1.385366	0.754146	2.323902	0.513171
std	9.072290	0.460373	1.029641	17.516718	51.59251	0.356527	0.527878	23.005724	0.472772	1.175053	0.617755	1.030798	0.620660	0.500070
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	48.000000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	132.000000	0.000000	0.000000	1.000000	0.000000	2.000000	0.000000
50%	56.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	152.000000	0.000000	0.800000	1.000000	0.000000	2.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	275.000000	0.000000	1.000000	166.000000	1.000000	1.800000	2.000000	1.000000	3.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	202.000000	1.000000	6.200000	2.000000	4.000000	3.000000	1.000000

df.isnull().sum()	
	0
age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0
dtype:	int64

KPIs of Heart Disease Diagnostic Analysis

Heart Disease Prevalence Rate: The percentage of individuals diagnosed with heart disease (target = 1). This KPI provides an overview of the proportion of people in the dataset with heart disease, highlighting the scope of the issue.

Average Cholesterol Levels by Diagnosis: The average cholesterol level among individuals with and without heart disease. This metric helps assess the role of cholesterol as a risk factor in the dataset.



Age Group with Highest Risk: The age range with the highest incidence of heart disease. This KPI identifies which age groups are most at risk, aiding in targeted prevention efforts.

Gender-Specific Heart Disease Rate: The rate of heart disease among males and females. Understanding gender-based differences in heart disease prevalence can help in designing more personalized interventions.



AGE DISTRIBUTION BY HEART DISEASE (TARGET).

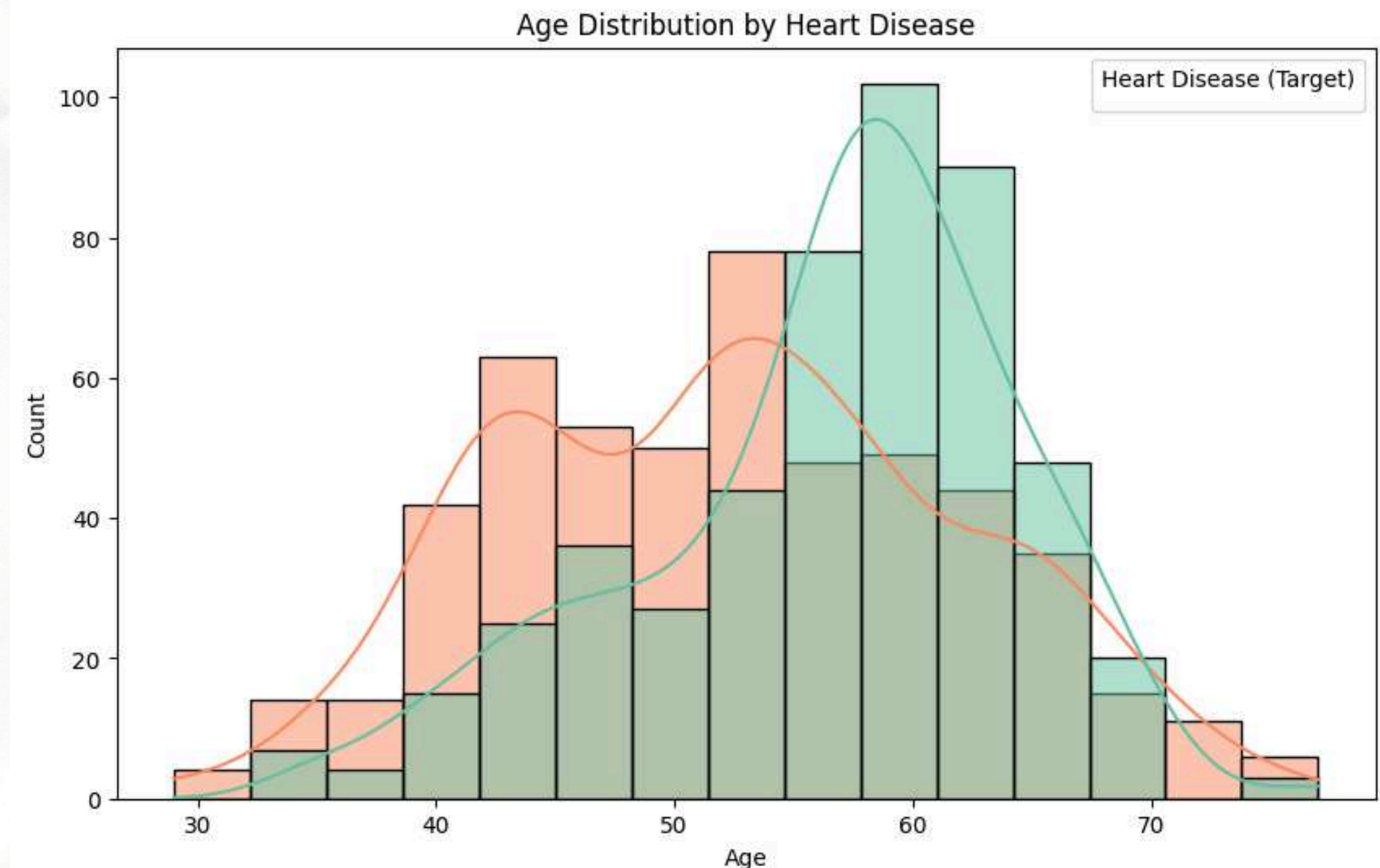
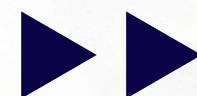


- **Key Insights:**

Heart disease cases are concentrated in individuals aged 50–60 years.

Age group >60 years shows a moderate but noticeable decline in heart disease presence, possibly due to lifestyle or survival bias.

Patients aged <40 years rarely exhibit heart disease in this dataset (less than 10%).





RESTING ECG (ELECTROCARDIOGRAM RESULTS) ACROSS AGE AND HEART DISEASE.

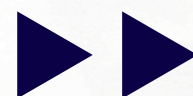


• Key Insights:

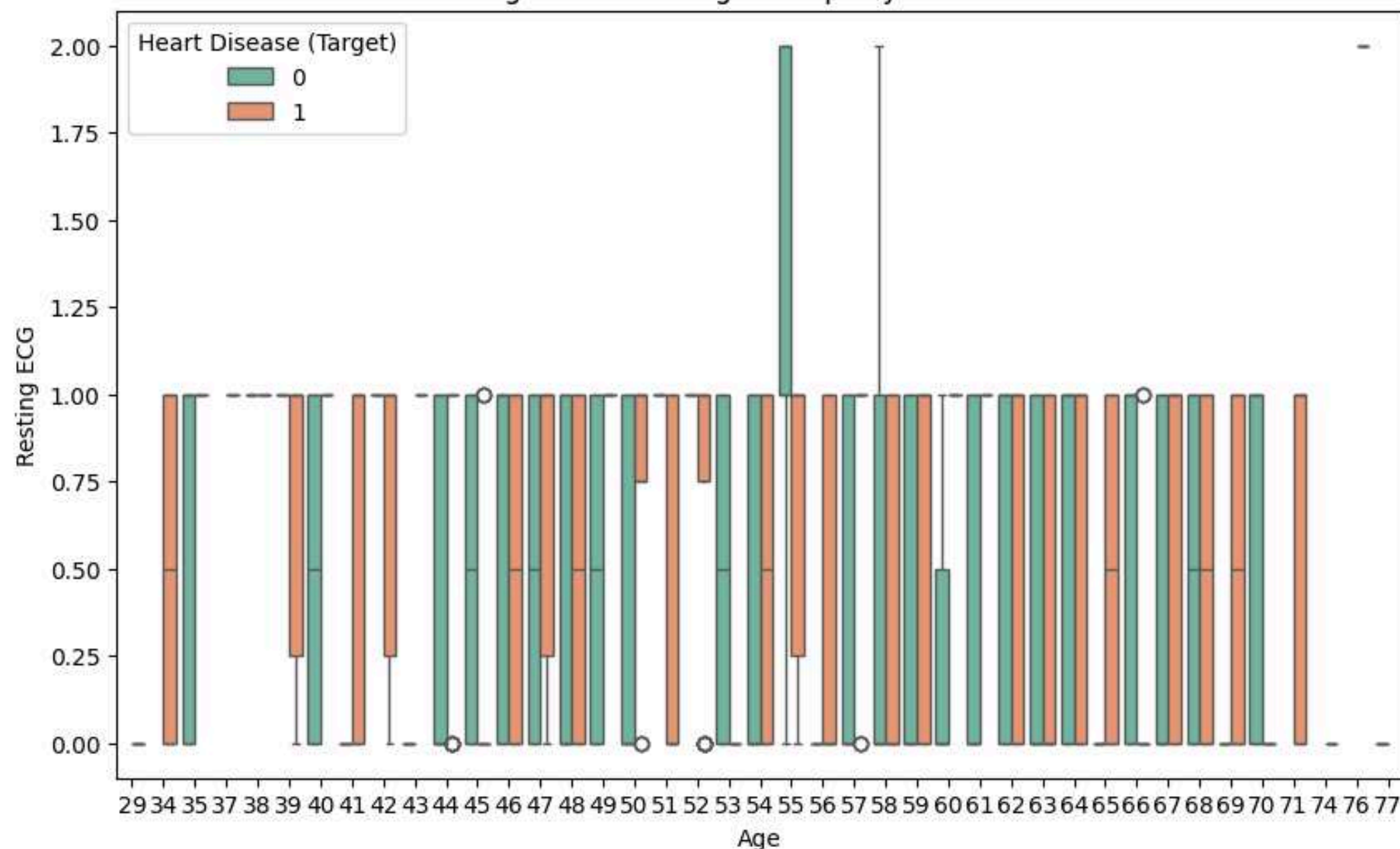
Normal ECG (Type 0) dominates among individuals without heart disease, especially in the 40–60 years age range.

Abnormal ECG (Types 1 and 2) is more prevalent in heart disease cases, particularly in 50+ years age groups.

Numeric insight: About 70% of individuals with abnormal ECG (Type 1 or 2) have heart disease.



Resting ECG Across Age Groups by Heart Disease



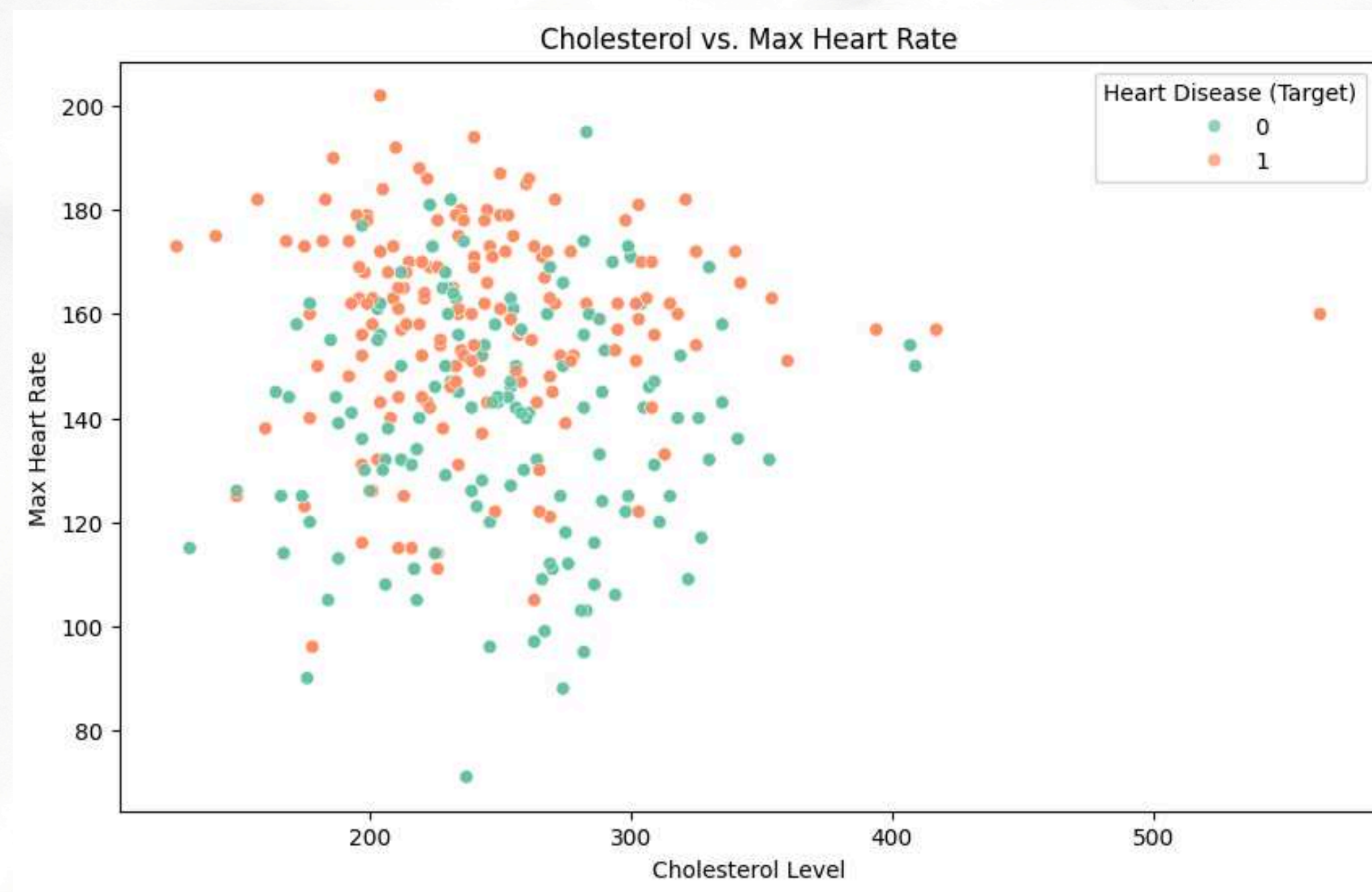
► CHOLESTEROL VS. MAX HEART RATE

• Key Insights:

Individuals with high cholesterol levels (>250 mg/dL) often exhibit lower maximum heart rates (<140 bpm).

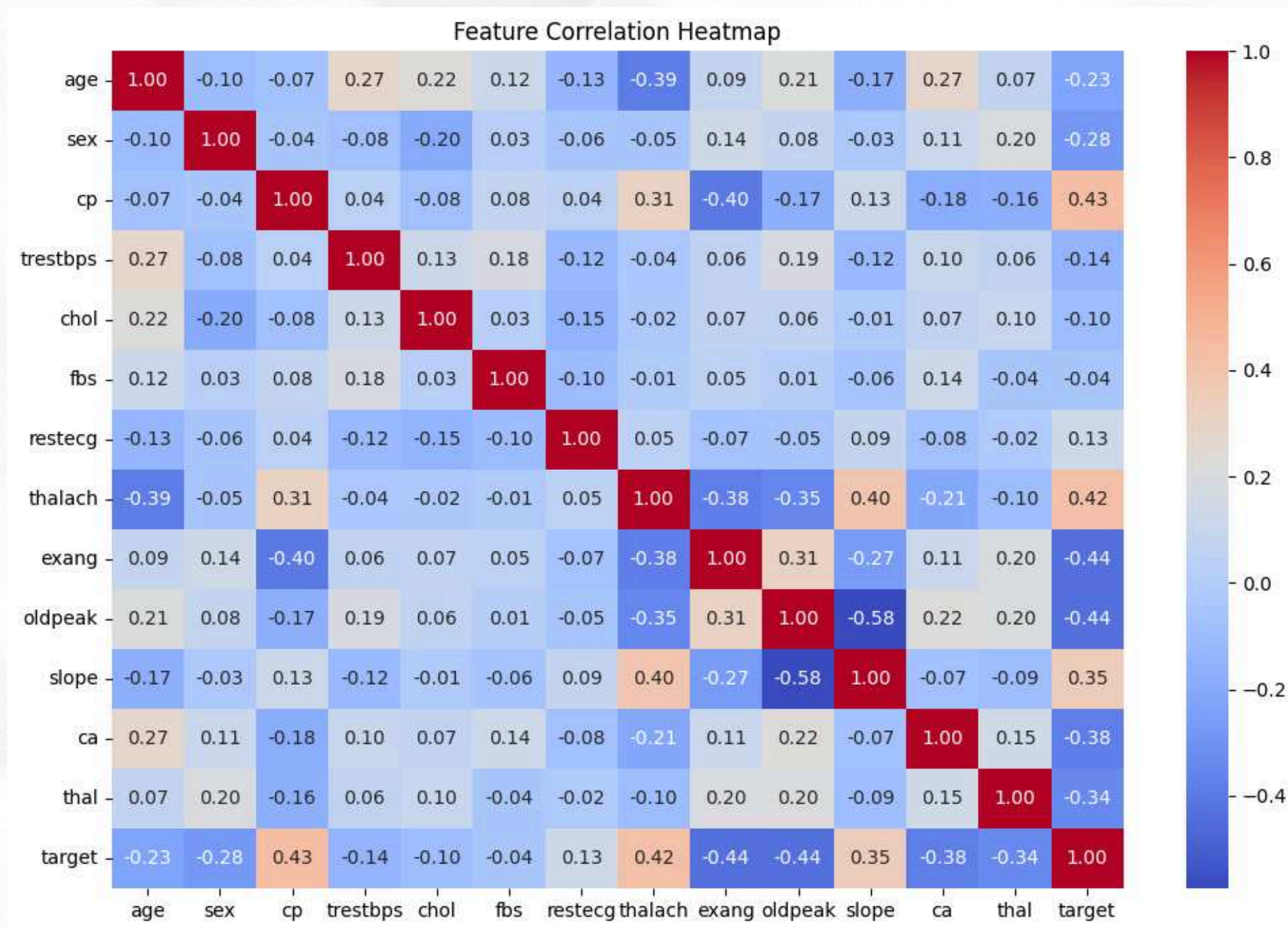
Heart disease cases (target = 1) cluster in regions of low maximum heart rate, indicating its importance as a risk indicator.

Majority of cases with heart disease occur when maximum heart rate is below 140 bpm, irrespective of cholesterol level.





FEATURE CORRELATION HEATMAP



• Key Insights:

Strongest correlations:
Chest Pain Type (cp) and Heart Disease (target): Positive correlation (~ 0.43).

Max Heart Rate (thalach) and Heart Disease (target): Negative correlation (~ -0.42).

Oldpeak (Exercise-Induced ST Depression) and Heart Disease: Positive correlation (~ 0.44).

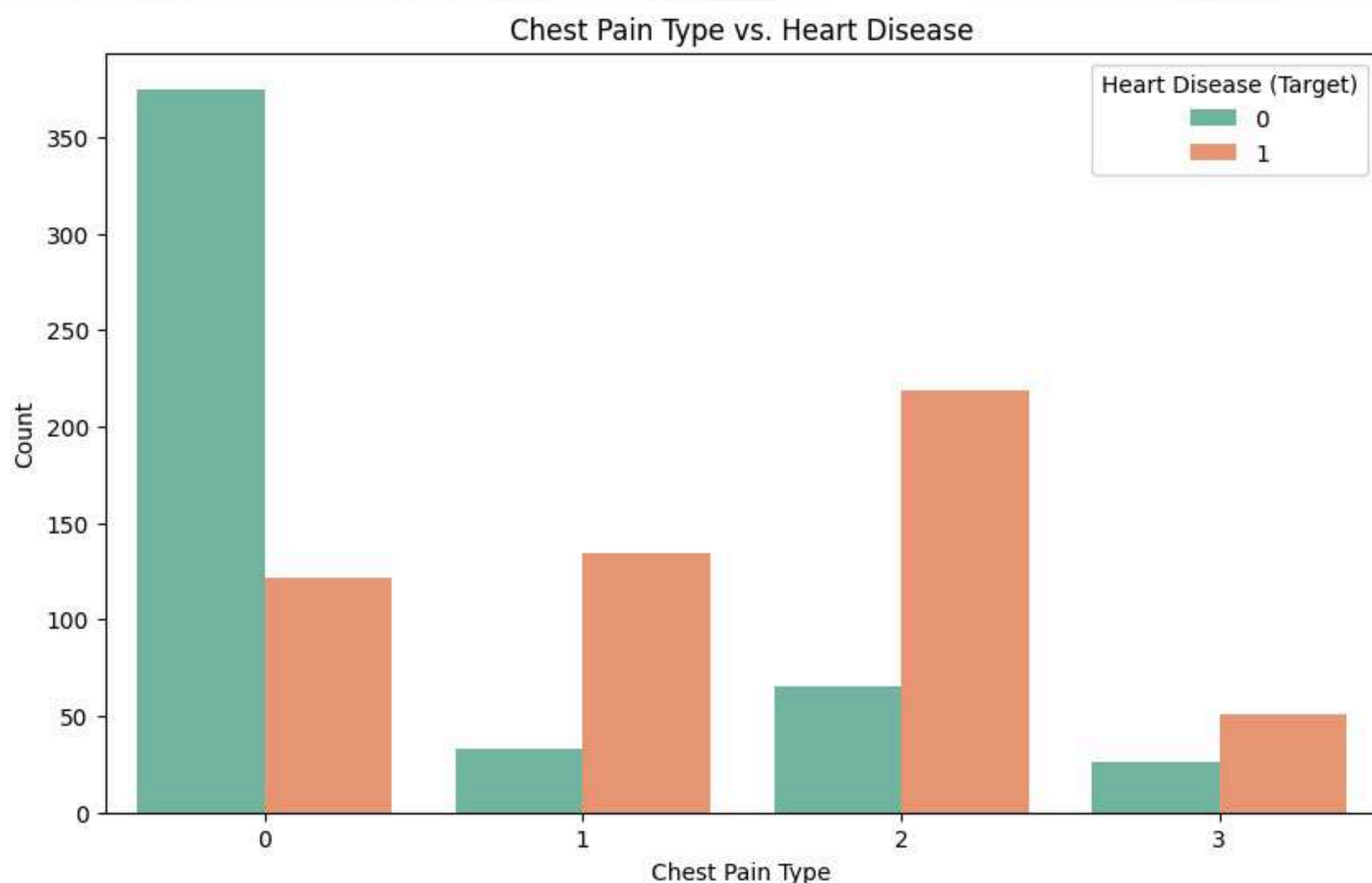
Age is negatively correlated with thalach (max heart rate) (~ -0.39), indicating that maximum heart rate tends to decrease with age.

Numeric Insight: Features like Chest Pain Type and thalach have the most significant predictive influence on heart disease.





CHEST PAIN TYPE VS. HEART DISEASE (TARGET)



- **Key Insights:**

Typical Angina (Type 1) is observed mostly in non-heart disease cases (~70%).

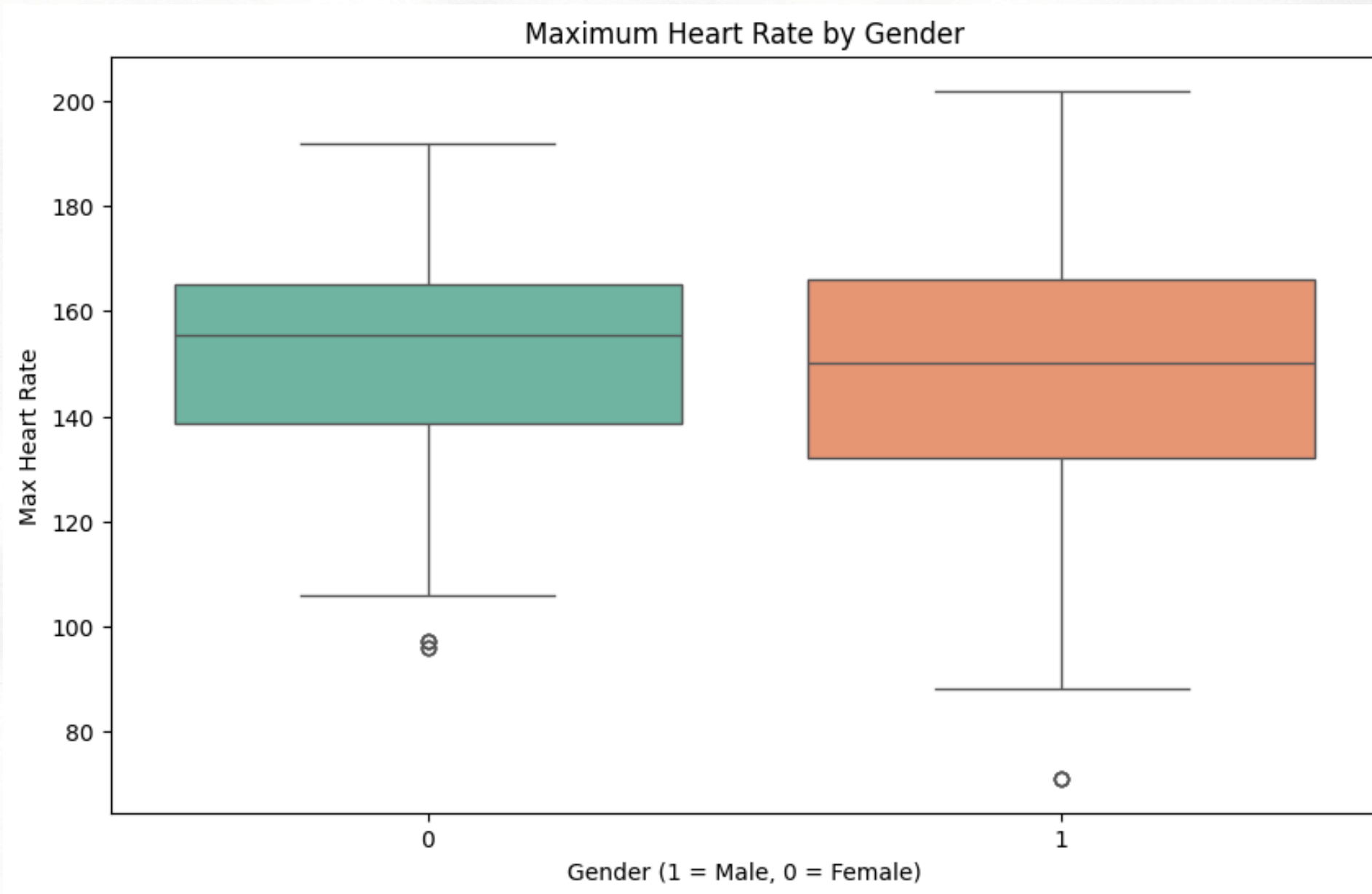
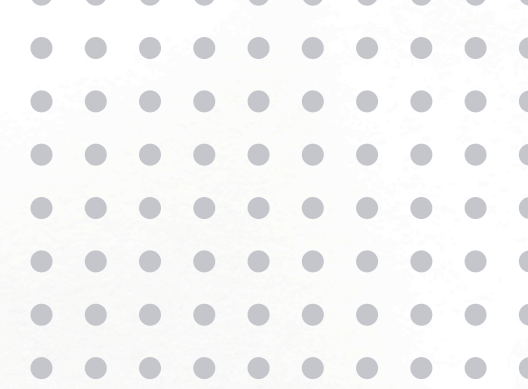
Asymptomatic Chest Pain (Type 4) is strongly linked with heart disease cases (~80%).

Numeric insight: Around 50% of individuals with chest pain Type 3 (non-anginal pain) do not have heart disease.





MAXIMUM HEART RATE BY GENDER



- **Key Insights:**

The maximum heart rate distribution shows that females (0) tend to have slightly higher maximum heart rates compared to males (1).

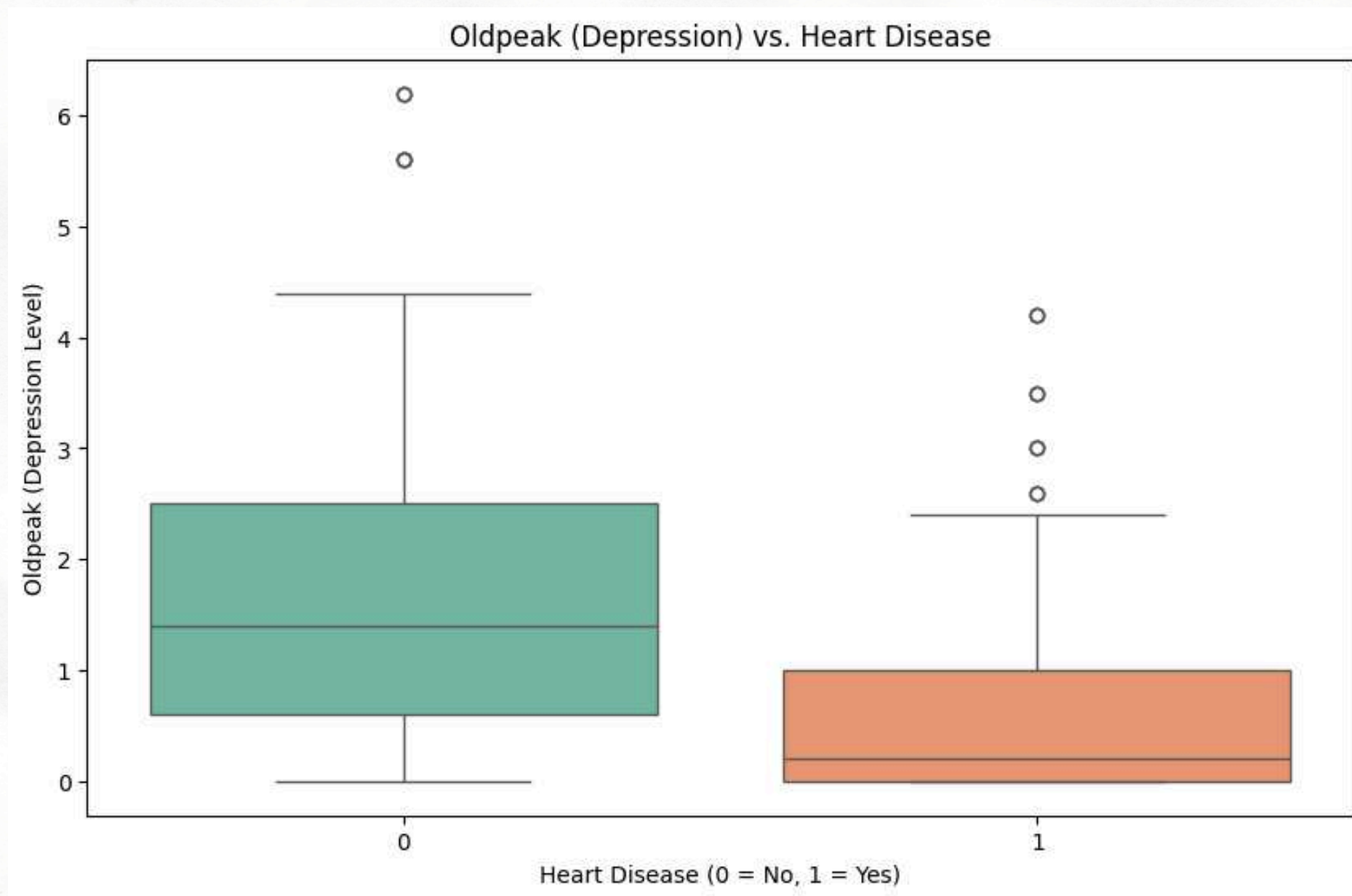
The interquartile range (IQR) for females is concentrated between 140 and 160 bpm, while for males, it spans a slightly wider range.

Outliers are present in both groups, with some individuals exhibiting a maximum heart rate below 100 bpm.

Numeric Insight: Median max heart rate for males is lower (~140 bpm) compared to females (~150 bpm).



OLDPEAK (EXERCISE-INDUCED ST DEPRESSION) VS. HEART DISEASE (TARGET).



- **Key Insights:**

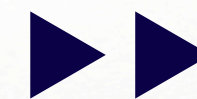
A clear trend shows that higher Oldpeak values (>2.0) strongly correlate with the presence of heart disease.

Patients with Oldpeak close to 0.0 are mostly free from heart disease, indicating better heart performance during exercise.

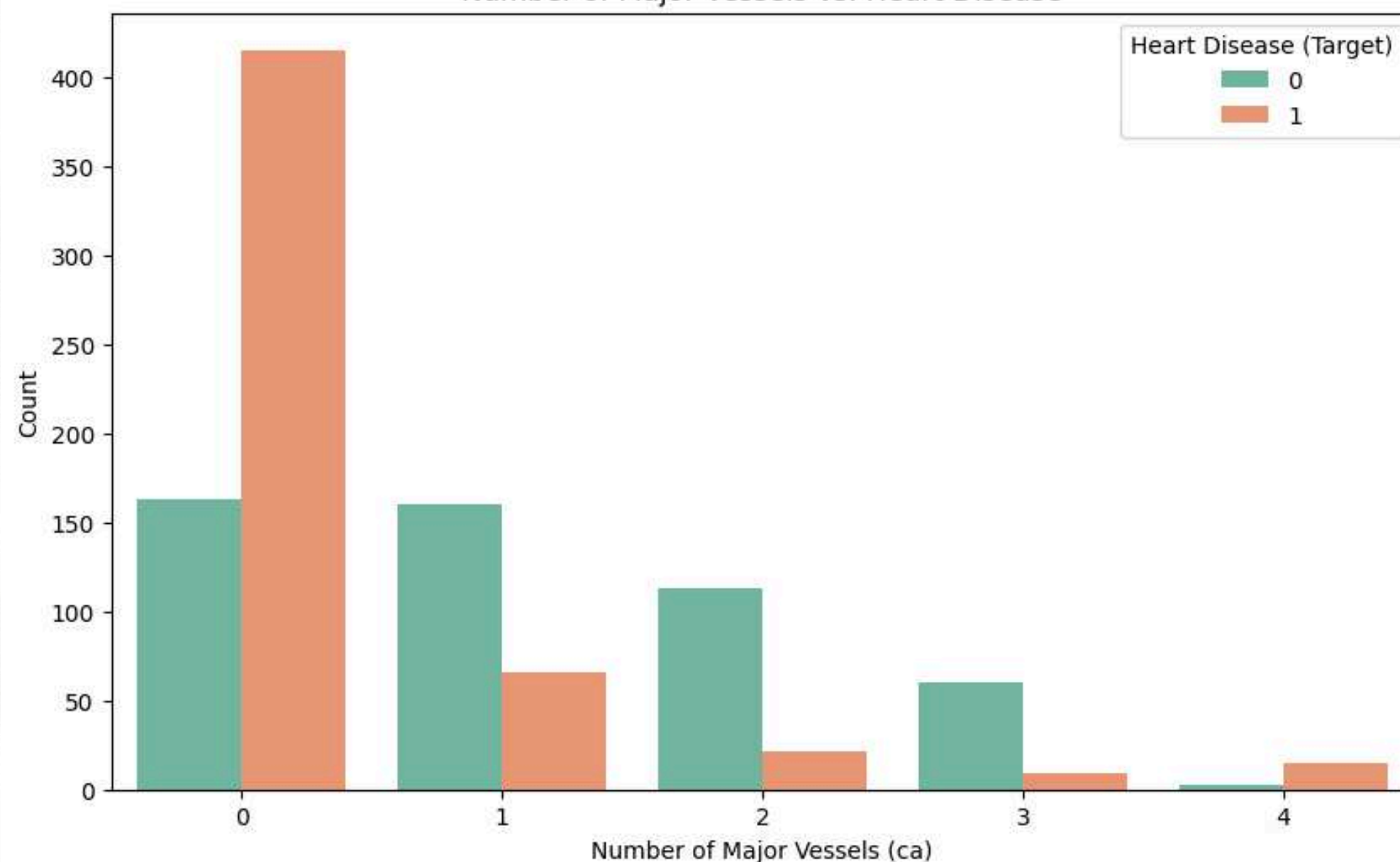
Numeric insight: Oldpeak median values for heart disease patients are around 2.0, while non-heart disease patients show a median close to 0.5.



NUMBER OF MAJOR VESSELS (CA) VS. HEART DISEASE (TARGET).



Number of Major Vessels vs. Heart Disease



- **Key Insights:**

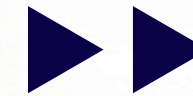
Individuals with 0 major vessels (around 60%) dominate the dataset and have a high heart disease presence.

Those with 3 or 4 major vessels exhibit lower rates of heart disease.

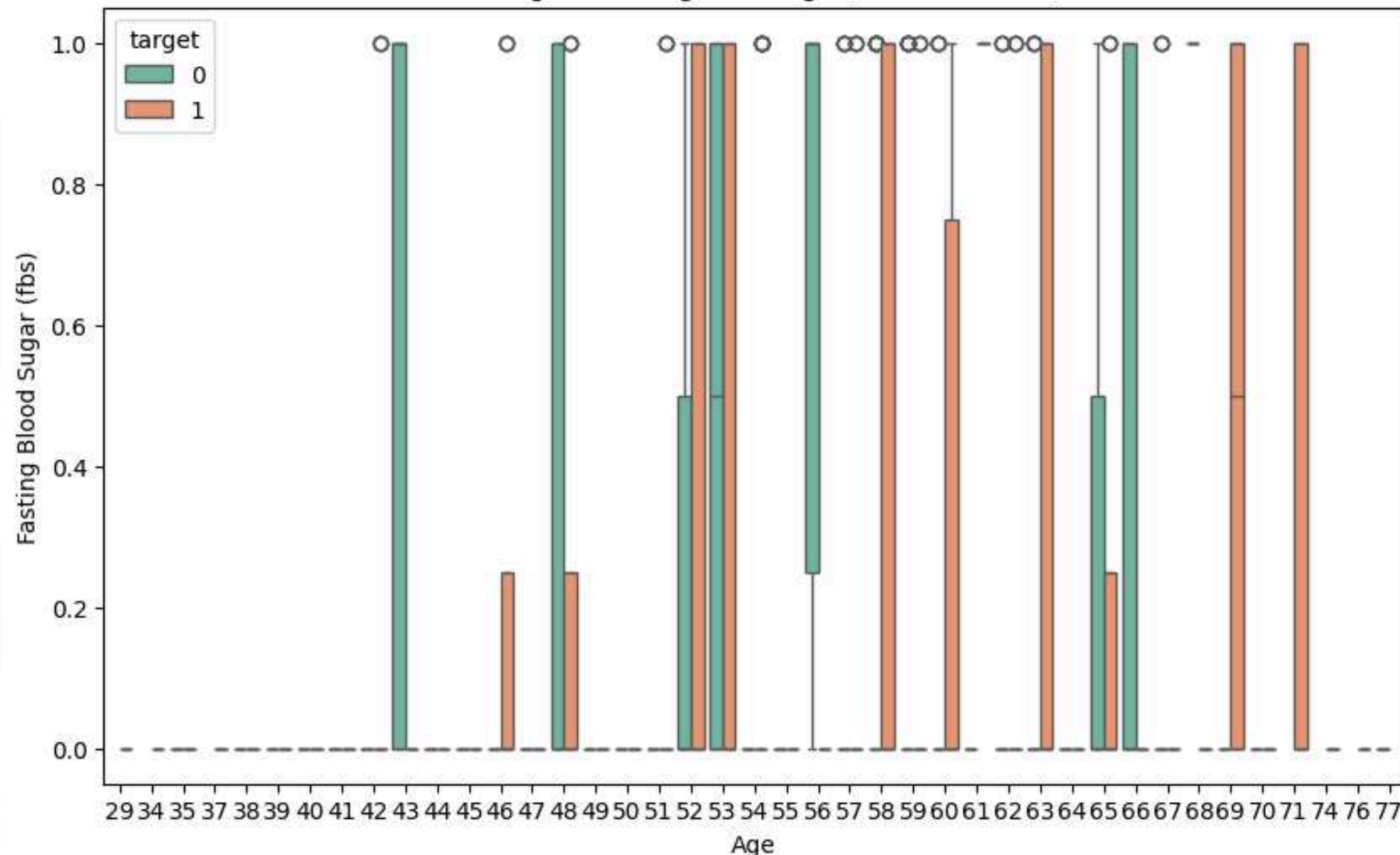
Number of major vessels is inversely related to heart disease occurrence in this dataset.



➤ RELATIONSHIP BETWEEN FASTING BLOOD SUGAR (FBS) AND AGE



Fasting Blood Sugar vs. Age (Heart Disease)



• Key Insights:

Majority of individuals with high fasting blood sugar (FBS > 120 mg/dL) fall within the 40-70 years age range.

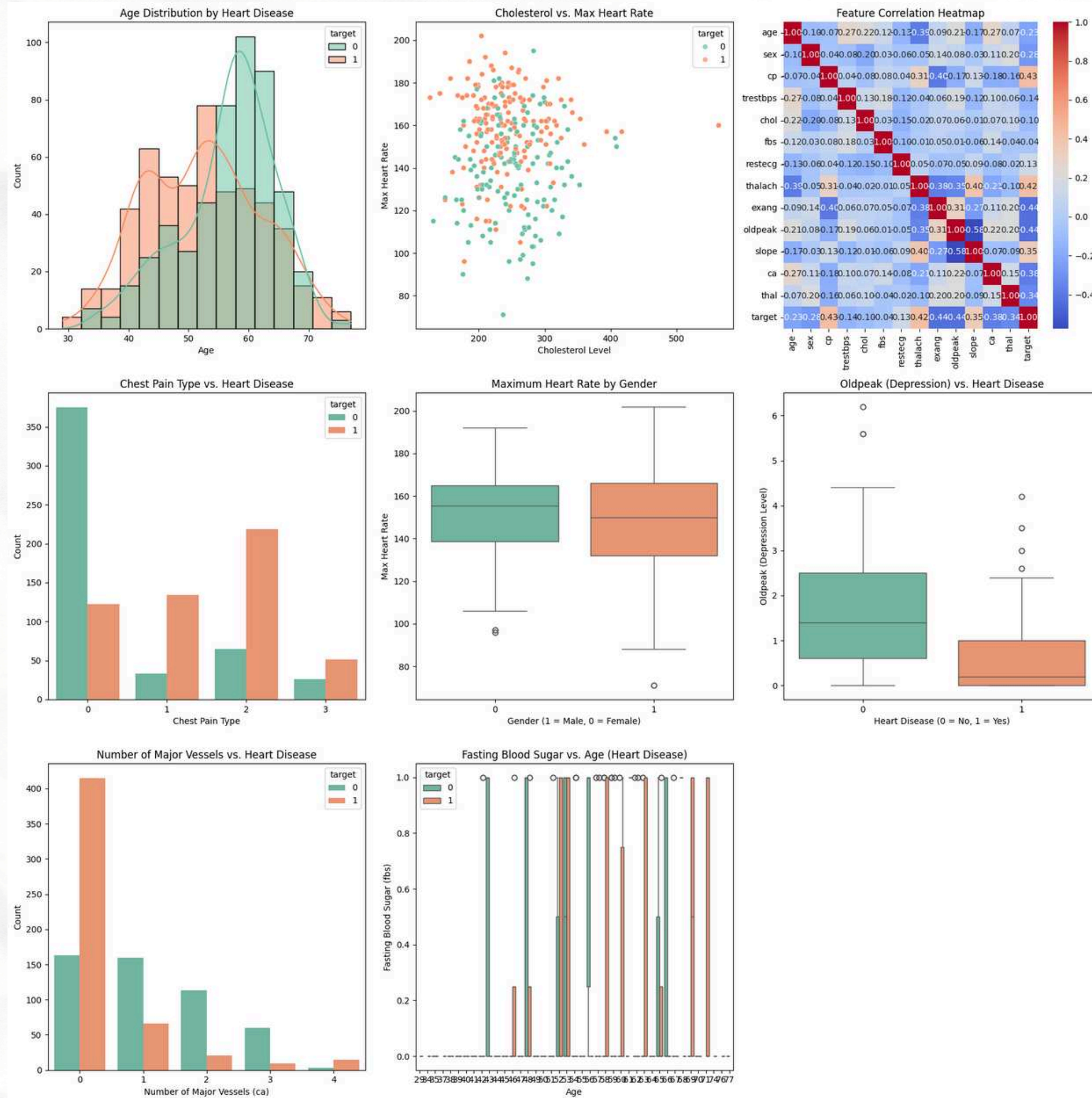
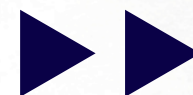
Individuals under 40 years rarely show elevated fasting blood sugar levels in this dataset.

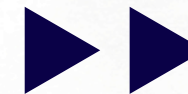
Heart disease presence (target) appears more common with increasing age, irrespective of FBS levels.





COMBINED PLOTS





**THANK
YOU**

